The power of servers, in serverless form

Enable Fluid Request demo

Fluid compute combines the efficiency of servers and the flexibility of serverless, enabling real-time, dynamic workloads like APIs, streaming, and AI.

Pay for compute, not idle time

Fluid runs many requests on a single instance. Billing is based on CPU usage.



Traditional serverless wastes idle time. Provisioned instances can sit idle between requests.



Fluid runs invocations concurrently. Overall cloud resources provisioned are dramatically reduced while invocations are kept alive in-memory.

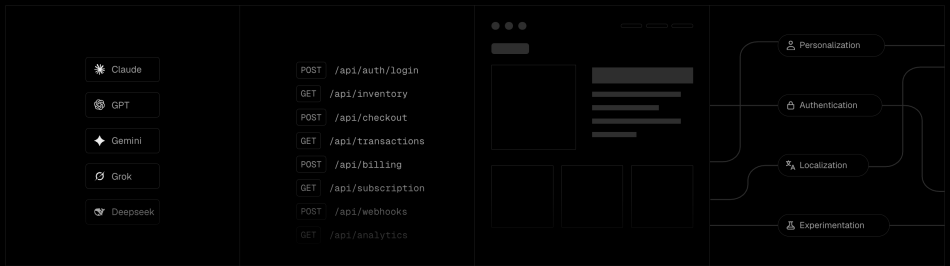


CPU that's charged when active, not idle. No more paying for I/O wait times, network latency, and API calls.

VERSE

> "By leveraging in-function concurrency, we were able to share compute resources between invocations, cutting costs by over 50% with zero code changes."

Lead Fullstack developer



AI workloads. Run tasks with reduced latency and higher concurrency, delivering faster, scalable results for all users, regardless of the workload size.

Business-critical APIs. Fluid Compute runs multiple API invocations concurrently on a single instance to handle traffic spikes.

Server-side and partial pre-rendering. Pre-render dynamic pages before requests arrive to reduce work at request time.

Middleware. Perform authentication checks and apply personalization with Fluid Compute.

Vercel Functions

Bridging servers and serverless

Taking the best of servers and serverless to create a new model in computing, scaling business-critical workloads efficiently across global environments.

FEATURES	SERVERS	SERVERLESS	FLUID
Cold start handling	Not applicable	Cold starts	Cold-start prevention
Scaling	Manual scaling	Auto-scaling	Efficient auto-scaling
Concurrency	Horizontal	Vertical	Horizontal & vertical
Operational overhead	High maintenance	Minimal, inefficient	Minimal, optimized
Pricing model	Upfront cost per server	Pay-as-you-go model	Pay-as-you-compute
CPU efficiency	High efficiency	I/O bound inefficiency	Optimized I/O efficiency

Fluid compute for dynamic web applications

In-function concurrency

Run multiple invocations on a single function instance, reducing idle compute time and lowering costs.

Cold-start reduction

Functions are pre-warmed and use bytecode caching to reduce cold starts.

Streaming

Send data to users as it becomes available, improving performance for AI, media, and real-time apps.

Cross-region failover

Reroutes traffic to backup regions during outages.

Dynamic scaling

Automatically adjusts concurrency and resource allocation based on real-time demand.

Post-response tasks

Keep functions running after sending a response to handle tasks like logging or database updates.

Experience fluid

Contact sales | Start deploying

Agent Stack
- AI SDK
- AI Gateway
- Sandbox
- Workflows
- Connect New
- Passport New
- eve New

Core Platform
- CI/CD
- Content Delivery
- Fluid Compute
- Observability

Security
- Platform Security
- WAF
- Bot Management
- BotID

Tools
- Vercel Drop
- Vercel Agent
- Vercel Plugin
- Agent Skills
- Domains
- v0

Frameworks
- eve New
- Next.js
- Nuxt
- SvelteKit
- Nitro
- Turborepo
- Tanstack Start
- FastAPI
- All frameworks

SDKs
- Vercel SDK
- Workflow SDK
- Flags SDK
- Chat SDK
- Queues SDK
- Streamdown

Build
- AI Apps
- Web Apps
- Marketing Sites
- Platforms
- Commerce
- Platform Engineers
- Design Engineers

Learn
- Docs
- Blog
- Changelog
- Knowledge Base
- Academy
- Articles
- Community
- Is Agentic

Explore
- Customers
- Marketplace
- Templates
- Partner Finder
- Vercel + AWS

Company
- About
- Careers
- Press
- Events
- Startups
- Shipped on Vercel
- Open Source Program
- Enterprise
- Pricing
- Help

Legal & Trust
- Privacy Policy
- Terms of Service
- Cookie Policy
- DPA
- Acceptable Use Policy
- Legal (all documents)
- Trust Center
- Status

Social
- GitHub
- X
- LinkedIn
- YouTube
- Instagram

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ALL SYSTEMS NORMAL.